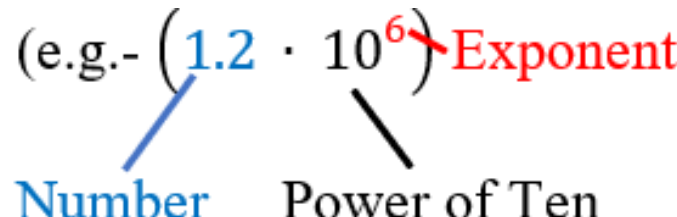


3.8 Writing Large Numbers in Scientific Notation

Lesson Introduction

 Benchmarks	MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency. MA.8.NSO.1.4 Express numbers in scientific notation to represent and approximate very large or very small quantities. Determine how many times larger or smaller one number is compared to a second number.		 Learning Targets	- I can use laws of exponents to evaluate numerical expressions with a base of 10 and exponents greater than or equal to 0. - I can express large numbers using scientific notation. - I can determine how many times larger or smaller a number is compared to another expressed in scientific notation.
 Benchmark Coherence	Building On MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole number exponents and rational number bases.		Working Toward MA.912.NSO.1.1 Extend previous understanding of the laws of exponents to include rational number exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.	
 Essential Questions	- How do you use the laws of exponents to evaluate numerical expressions with a base of 10? - How do you write a large number using scientific notation? - How do you write the standard form of a number expressed in scientific notation? - How many times larger or smaller is the number expressed in scientific notation than the other number expressed in scientific notation?			
 Lesson Narrative	Students extend their prior knowledge of the laws of exponents to evaluate numerical expressions with bases of 10 and natural number exponents. Students then discover how to write large numbers in scientific notation. They also determine how many times larger or smaller a number written in scientific notation is than another number written in scientific notation. Throughout the lesson students see real-world examples of large numbers and express them in scientific notation.			
 Component Summary	3.8.1 Warm-Up (~5 min.)	Explore applications of percentages and write large numbers (<i>SE p. 140</i>)	3.8.4 Guided Instruction (10 min.)	Write large numbers in scientific notation (<i>SE p. 142</i>)
	3.8.2 Exploration (10 min.)	Explore powers of ten and place value (<i>SE p. 140</i>)	3.8.5 Collaboration (10 min.)	Compare large numbers written in scientific notation (<i>SE p. 144</i>)
	3.8.3 Exploration (10 min.)	Explore patterns in scientific notation (<i>SE p. 141</i>)	3.8.6 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 145</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> Scientific Notation <u>Additional Terms:</u> N/A		(Optional) Highlighters or Colored Pencils - Calculator	- Consider creating scientific notation templates for students to fill in when expressing numbers in scientific notation. - Consider adding the laws of exponents to the classroom Word Wall or creating Anchor Charts.

 Teacher Tips	Common Misconceptions - Students may incorrectly think that the difference in the powers of 10 is how many times larger or smaller a number is than another number. For example, they may think 10^5 is two times larger than 10^3, because $5 - 3 = 2$, instead of 10^2 or 100 times larger. - Students may count the one, instead of only the zeros, when they express the numbers using a power of 10.	Things to Consider - Students may be overwhelmed by the large numbers. Discuss that this is why we use exponents to express the numbers. - Consider creating an anchor chart with the form of scientific notation as $a \times 10^n$ where $1 \leq a < 10$. Numbers can be expressed using powers of 10 that are not in this form, but the definition of scientific notation is that the absolute value of a should be at least 1 and less than 10. Consider using both the mathematical representation of the constraint on a as well as the written description of the constraint.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Turn and Talk Prompt students to take turns during 3.8.5 Collaboration. Have students to share their strategies and answers. Then, ask the partner to share if they agree or disagree and why.	MA.K12.MTR.2.1: Multiple Representations In 3.8.3 Exploration students explore and discover how to use multiple representations to express large numbers. Engage students in discussions about how the scientific notation is connected to the standard form of the number.
 Differentiate Instruction	In 3.8.4 Guided Instruction, provide a visual entry point by providing students with a template to use when expressing numbers in scientific notation. Students can also create a Venn diagram to record the similarities and differences they notice between the scientific notation and standard form of the values.	
		
Lesson Notes:		

3.8.1 Warm-Up: Grains of Sand

Component Overview: Students recognize that the same ratio of quartz in the sand on the beach will exist in a teaspoon of sand. They also practice writing the standard form of a large number.



3.8.1 Warm-Up: Grains of Sand

The beaches on Siesta Key are 99% quartz, meaning that the beaches are made of some of the finest grains of sand.



- Suppose you have a teaspoon of the sand from Siesta Key. How could the teaspoon of sand be used to estimate the number of grains of sand in a bucket of sand?

Answers will vary. If you count the number of grains of sand in a teaspoon, and then figure out how many teaspoons are in the bucket, you can multiply the number of grains in a teaspoon by this number to determine the number of grains in the whole bucket.

A group of researchers at the University of Hawaii used a teaspoon of sand to estimate the number of grains of sand on all of the beaches on the Earth. Their estimate was that there are roughly seven quintillion, five hundred quadrillion grains of sand. A quintillion is 1,000,000,000,000,000 and a quadrillion is 1,000,000,000,000.

- Write seven quintillion, five hundred quadrillion.
7,500,000,000,000,000



Florida has a wide range of sands. Consider using this opportunity to have students share their life experiences with the different types of sands – from beaches to their backyards. This may give students an opportunity to share something of interest to them.

3.8.2 Exploration: Powers of Ten

Component Overview: Students explore the relationship between powers of 10, their values, and the place value of the one. Students discover that the exponent indicates the number of zeros in the value.



3.8.2 Exploration: Powers of Ten

1. Use a scientific calculator to find the value of each power of 10.

Power of 10	Value
10^1	10
10^2	100
10^3	1000
10^4	10,000
10^5	100,000
10^6	1,000,000
10^7	10,000,000
10^8	100,000,000
10^9	1,000,000,000
10^{10}	10,000,000,000

2. What do you notice about the relationship between the power of 10 and its value?

Answers will vary. The 1 is the same number of places from the left as the value of the exponent.

3. Work with your partner to complete the table.

Power of 10	Value	Place Value of 1
10^1	10	Tens
10^2	100	Hundreds
10^3	1000	Thousands
10^4	10,000	Ten Thousands
10^5	100,000	Hundred Thousands
10^6	1,000,000	Millions

4. Work with your partner to describe how the exponent of the power of ten relates to place value.

Answers will vary. There are the same number of zeroes as the exponent.

5. Based on the pattern in the table, what is the value of 10^0 ?

1



Exploring the laws of exponents with powers of 10 will allow students to work with relationships of place value to solidify their understanding of the laws of exponents.



Review place values and have students create a reference sheet or foldable in their interactive notebooks. Ask students to share how they remember the place values.

3.8.3 Exploration: Patterns in Scientific Notation

Component Overview: Students explore patterns in scientific notation. They discover that the exponent is equal to the number of zeros in the numerical value and that the exponent is the same in the power of ten and in the scientific notation. Finally, students consider real-world quantities that could be represented by the large numbers.



3.8.3 Exploration: Patterns in Scientific Notation

Very large numbers can be written using powers of 10 in a shorthand method called scientific notation.

1. Work with your partner to complete the table.

Power of 10	Value	Scientific Notation	Word Form
10^2	100	1.0×10^2	one hundred
10^3	1,000	1.0×10^3	one thousand
10^4	10,000	1.0×10^4	ten thousand
10^5	100,000	1.0×10^5	hundred thousand
10^6	1,000,000	1.0×10^6	one million
10^9	1,000,000,000	1.0×10^9	one billion
10^{12}	1,000,000,000,000	1.0×10^{12}	one trillion

2. Select two of the numbers. Work with your partner to name something in the world that is described by those numbers.

Answers will vary.

Number	What can be described by this number in the real world?
100	years in a century
1000	millimeters in a meter

3. Work with your partner to describe at least two patterns in the relationships among the power of 10, the numerical value, the scientific notation, and/or the word form.

Answers will vary.

- a. The exponent is the same in the power of ten and in the scientific notation.
- b. The number of zeros in the numerical value is the same as the exponent in the power of ten.



Students explore and discover how to use multiple representations to express large numbers. Engage students in discussions about how the scientific notation is connected to the standard form of the number.



If students finish early, prompt them to list a real-world example for each number in the table. Have a variety of students share their examples so students have the opportunity to see the utility of the lesson learning targets.

3.8.4 Guided Instruction: Writing Large Numbers in Scientific Notation

Component Overview: Students discover how to write large numbers in scientific notation. Students also reflect on the similarities and differences between the standard form and scientific notation of large numbers.



3.8.4 Guided Instruction: Writing Large Numbers in Scientific Notation

- Consider the number 2,000,000.
 - What power of ten represents 1 million? $10^{\boxed{6}}$
 - Complete the multiplication sentence to express 2,000,000 using this power of 10.

$$\boxed{2} \times 10^{\boxed{6}} = 2,000,000$$



Scientific notation is a method of writing very large or very small numbers using exponents in which a number is expressed as the product of a power of 10 and a number that is at least 1 and less than 10.

- Complete the statement.
When rewriting 2,000,000 in scientific notation, the factor

2 is ☐ less than ☒ greater than or equal to 1 and

☐ less than ☒ greater than or equal to 10. The digit 2 is in the

millions place. A million expressed as a power of 10 is

☐ 10^5 . ☒ 10^6 . Therefore, 2,000,000 is written as ☐ 2×10^5 ☒ 2×10^6

in scientific notation.

- The expression 20×10^5 is also equivalent to 2,000,000. Use the definition of scientific notation to explain why 20×10^5 is not the correct expression of 2,000,000 in scientific notation.

The factor multiplied by the power of ten must be between 1 and 10.



Rewriting numbers in scientific notation should not be taught by moving decimal places, but rather in the context of place value and powers of 10 as built from 3.8.2 Exploration. The Guided Instruction is set up to continue using place value.



Provide a visual entry point by providing students with a template that uses place value to use when expressing numbers in scientific notation.



Consider adding the term “Scientific notation” to the classroom Word Wall for students to refer to. Discuss when students have seen scientific notation in their everyday lives and why scientific notation is beneficial.

3.8.4 Guided Instruction: Writing Large Numbers in Scientific Notation (continued)

4. Consider the standard form and scientific notation of the numbers shown in the table.

Standard Form	Scientific Notation
2,300,000	2.3×10^6
2,030,000	2.03×10^6

Describe the similarities and differences between the values expressed in scientific notation.

Similarities	Differences
- Both have a factor being multiplied by 10^6. - Both of the original numbers have a 2 in the millions place. - Both of the original numbers are to the millions. - The factor being multiplied by 10^6 is between 1 and 10.	- The 3 is in a different place in the two numbers in standard form. - The factor being multiplied by 10^6 is not the same.

As explored in the previous lesson, the powers of 10 relate directly to place values.

5. The planet Venus is about 26,000,000 miles from earth.
- What value between 1 and 10 could be multiplied by a power of 10 to result in 26,000,000?
 2.6
 - Which digit is in the ones place of your answer to Part A?
 2
 - In which place value is that digit in 26,000,000?
 ten million
 - Which power of 10 relates to this place value?
 10^7
 - Express 26,000,000 in scientific notation.
 2.6×10^7
6. In 2018, funding for education in Florida totaled \$30,100,000,000. Express this value in scientific notation.
 3.01×10^{10}



Spend time discussing the difference between the two values of a in question 4. Focus students' attention to the zero in 2.03 and that there is no zero in 2.3.

Point students to the anchor chart that shows that the number, a , in $a \times 10^n$ must fit the constraint $1 \leq |a| \leq 10$ to be considered scientific notation. Numbers written using powers of 10 where the number a does not fit this constraint are, by definition, not considered to be written in scientific notation.



Students can create a Venn diagram to record the similarities and differences they notice between the scientific notation and standard form of the values.



Students may struggle determining the value of a . Refer students back to question 4 for an example.

3.8.5 Collaboration: Comparing Large Numbers Written in Scientific Notation

Component Overview: Students compare large numbers written in scientific notation. They discuss how many times bigger a number written in scientific notation is than another number also written in scientific notation.



3.8.5 Collaboration: Comparing Large Numbers Written in Scientific Notation

1. Work with your partner to complete the following.

- a. Complete the table.

Scientific Notation	Standard Form
2×10^2	200
2×10^3	2,000
2×10^4	20,000
2×10^5	200,000

- b. Complete each statement.

2×10^3 is ☐ 10 ☐ 100 ☐ 1000 times ☐ smaller ☐ larger than 2×10^2 .

2×10^2 is ☐ 10 ☐ 100 ☐ 1000 times ☐ smaller ☐ larger than 2×10^4 .

2×10^5 is ☐ 10 ☐ 100 ☐ 1000 times ☐ smaller ☐ larger than 2×10^2 .

- c. Describe how to compare numbers written in scientific notation if the factors being multiplied by a power of 10 are the same.

You can compare the powers of ten.

2. Complete the table.

Scientific Notation	Standard Form
3.2×10^7	32,000,000
6.4×10^7	64,000,000
9.6×10^7	96,000,000

- a. How many times larger is 6.4×10^7 than 3.2×10^7 ?

It is two times larger.

- b. How many times smaller is 3.2×10^7 than 9.6×10^7 ?

It is three times smaller.



Students may incorrectly think that the difference in the powers of 10 is how many times larger or smaller a number is than another number. For example, they may think 10^5 is two times larger than 10^3 , because $5 - 3 = 2$. Remind students that it is 10^2 or 100 times larger.

3.8.5 Collaboration: Comparing Large Numbers Written in Scientific Notation (continued)

- c. Describe how to compare numbers written in scientific notation if the powers of 10 are the same.
You should compare the factors being multiplied by the power of ten.

3. Complete the table.

Scientific Notation	Standard Form
2.1×10^4	21,000
4.2×10^5	420,000
8.4×10^6	8,400,000

- a. How many times larger is 4.2 than 2.1? 2
- b. How many times larger is 10^5 than 10^4 ? 10
- c. How many times larger is 4.2×10^5 than 2.1×10^4 ?
 2 $\times 10$ = 20 times larger

d. Complete the statement.

The value of 8.4×10^6 is
☐ 4
☐ 40
☒ 400
 times larger than
 the value of 2.1×10^4 because 8.4 is
☐ 4
☐ 40
☒ 400
 times
 larger than 2.1, 10^6 is
☐ 10
☒ 10^2
☐ 10^3
 times larger than
 10^4 , and $\times$ =
☐ 4.
☐ 40.
☒ 400.

If the factors being multiplied by the powers of 10 are different, but one is not a multiple of the other, the comparison of the two values can be *approximated*.



Take Turns

Prompt students to take turns during 3.8.5 Collaboration. Have students to share their strategies and answers. Then, ask the partner to share if they agree or disagree and why.

3.8.5 Collaboration: Comparing Large Numbers Written in Scientific Notation (continued)

4. The average distance from Earth to the moon is 2.38855×10^5 miles. The average distance from Earth to the sun is about 9.3×10^7 .

Complete the statements to approximate how many times farther the sun is from Earth than the moon.

9.3 is about ☐ 2 ☒ 4 ☐ 6 times ☐ smaller ☒ larger than 2.38855 and

10^7 is ☐ 10 ☒ 10^2 ☐ 10^3 times ☐ smaller ☒ larger than 10^5 , so the value

of 9.3×10^7 is about 400 times larger than 2.38855×10^5 .



Consider reading the text-dense problems aloud in this and other components to provide an auditory entry point.



Consider having students multiply 400 times 2.38855×10^5 to verify their answer.

3.8.6 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



3.8.6 Wrap-Up: Cool Down

1. Explain how to express large numbers using scientific notation in your own words.

Answers will vary.



Consider providing sentence stems or answer options.